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Paleozoic Paleomagnetism and Northward Drift of the Alexander Terrane, Southeastern Alaska

ROB VAN DER VOO AND MERIDEE JONES

Department of Geology and Mineralogy, University of Michigan, Ann Arbor, Michigan 48109

C. SHERMAN GROMME, G. DONALD EBERLEIN, AND MICHAEL CHURKIN, JR.

U.S. Geological Survey, Menlo Park, California 94025

Paleozoic limestone, graywacke, sandstone, mudstone, red beds and volcanic rocks of the Alexander terrane, southeastern Alaska, have yielded six paleomagnetic pole positions after thermal and alternating-field demagnetization. These poles are from sample groups of late Middle Ordovician, Late Ordovician, Devonian, Late Devonian, and early and late Carboniferous age. To test various tectonic models for the structural development of this part of western North America, the paleomagnetic results are compared to those for the North American craton. It is found that the observed inclination and declination values deviate significantly from the values predicted for the present-day position of the Alexander terrane (55.5N, 133.5W). Better matching can be obtained for a paleoposition of the terrane at about 40N, 120W, in the present position of western Nevada and northeastern California. In addition, an in situ 25° clockwise rotation of the terrane is required to restore it to its original position.

INTRODUCTION

Paleozoic rocks of the Alexander terrane [*Berg et al.*, 1972, 1978] as well as of other terranes in the westernmost part of northern North America all differ from Paleozoic rocks of the eastern part of the Cordillera and therefore appear to have moved with respect to the stable North American craton. The likelihood of very large scale crustal movements in this region was expressed in plate-tectonic terms by *Monger and Ross* [1971], who used the occurrences of late Paleozoic fusulinaceans to divide the western Canadian Cordillera into three northwest-trending subparallel belts. The eastern and western belts both contain fusilinid assemblages of the North American (non-Tethyan) type dominated by Schwagerinidae and containing a few other fusulinids, whereas the central belt contains mainly Verbeekinidae accompanied by an assemblage of other fusilinids of Tethyan affinity. Monger and Ross showed that the late Paleozoic clastic and volcanic rocks of the eastern and western belts are rather similar but differ significantly from those of the central belt, and mentioned that blueschist-facies metamorphic rocks occur locally between the central and eastern belts.

Two alternative plate-tectonic models to account for the juxtaposition of these three belts were also offered by *Monger and Ross* [1971]. In the first model, the eastern and western belts formed above subduction zones of opposite polarity while the central belt formed near the intervening spreading center; the three belts then collapsed together with the cessation of spreading and accelerated subduction of the oceanic crust. The second model invokes large-scale strike-slip faulting whereby either the western belt alone or the central and western belts successively were transported northward and attached to the craton.

The western belt, of which the Alexander terrane forms a part, is a mixture of continental and oceanic rock types that has also been referred to as the insular belt and the coast plutonic complex by *Monger et al.* [1972] and as the borderland terranes or outer eugeosynclinal belt by *Churkin and Eberlein* [1977]. The Alexander terrane, first designated by *Berg et al.* [1972], is composed largely of submarine volcanic rocks and closely associated plutonic rocks, volcanically derived sedimentary rocks, and reef limestone, which taken together indicate a volcanic-arc origin. Although subsequent geologic events have obscured evidence of the original relationships, other Paleozoic rocks to the east of the Alexander terrane are characterized by ocean-floor volcanic and sedimentary rocks that have been interpreted as having been deposited directly on oceanic crust which is now found as associated mafic and ultramafic rocks [*Jones et al.*, 1972; *Monger et al.*, 1972; *Churkin and Eberlein*, 1977].

More recently, *Berg et al.* [1978] subdivided rocks approximately corresponding to the western belt of *Monger and Ross* [1971] into a number of tectonostratigraphic terranes and subterranes. Outboard (west) of the Alexander terrane are Wrangellia [*Jones et al.*, 1977; *Hillhouse*, 1977*a*], containing late Paleozoic and Mesozoic rocks, and the Chugach terrane, composed of Jurassic and Cretaceous rocks. *Berg et al.* [1978] divided the Alexander terrane into three subterranes named Annette, Admiralty, and Craig. The largest of these is the Craig subterrane, which is split into two parts by the Chatham Strait fault. The general configuration of the subterranes is shown in Figure 1 of the companion paper [*Hillhouse and Gromme*, 1980]. *Berg et al.* [1978] delineated four other tectonostratigraphic terranes that are included in the western belt of *Monger and Ross* [1971]. We will not discuss these eastern terranes further except to say that their existence implies several episodes of continental accretion in addition to the time during which the Alexander terrane presumably arrived at approximately its present position.

The inferred mobile nature of the western belt has engendered three hypotheses as to its place of origin. As outlined by *Churkin and Eberlein* [1977], the western belt could consist of (1) fragments of Asia [*Wilson*, 1968; *Danner*, 1970] or of a hypothetical now-dispersed continent [*Moores*, 1970] that has been called Pacifica by *Nur and Ben-Avraham* [1977], (2) volcanic arcs and para-autochthonous microcontinental frag-

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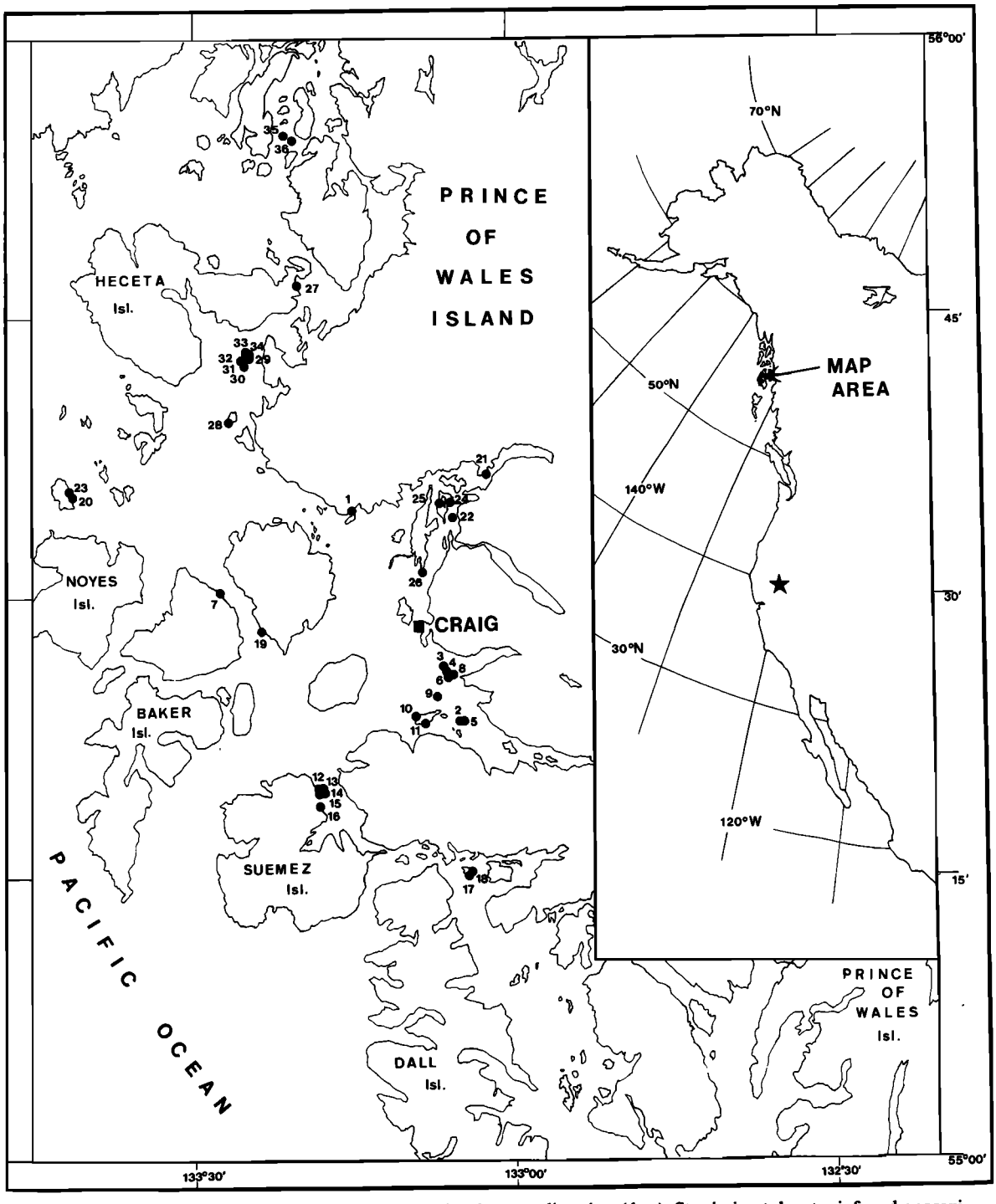


Fig. 1. Index map of part of southeastern Alaska, showing sampling sites (dots). Star in inset denotes inferred approximate original location of Alexander terrane.

ments that moved in an accordion-like fashion as marginal basins opened and closed behind them [*Churkin and Eberlein,* 1977], or (3) fragments of the North American continent or offshore island arcs displaced northward by large-scale strike-slip faulting [*Monger and Ross,* 1971]. The third hypothesis was expanded further for the Alexander terrane by *Jones et al.* [1972]. On the basis of Silurian rock types and facies trends in California and the Alexander terrane, Jones et al. suggested that the Alexander terrane originated as part of California and has since been carried northward to its present position by large-scale lateral shear. Such lateral shear would be observed primarily as latitudinal displacement because the great-circle distance between California and southeast Alaska coincides, for most of the Phanerozoic, with a paleomeridian. Paleomagnetically this displacement would be apparent as a discrepancy between the inclination recorded in rocks of the study area and that predicted by contemporaneous poles from the stable North American craton. If the study area had been rotated locally as well, or if the pole of rotation for the lateral shearing were much less than 90° away, a deviation between observed and predicted declination would also be observed. On the other hand, an accordion-like opening and closing of marginal seas, if unaccompanied by lateral shear, would imply motions principally along parallels of paleolatitude for most of geologic time. The paleomagnetic record would then agree in inclination, but the resulting pole position from the

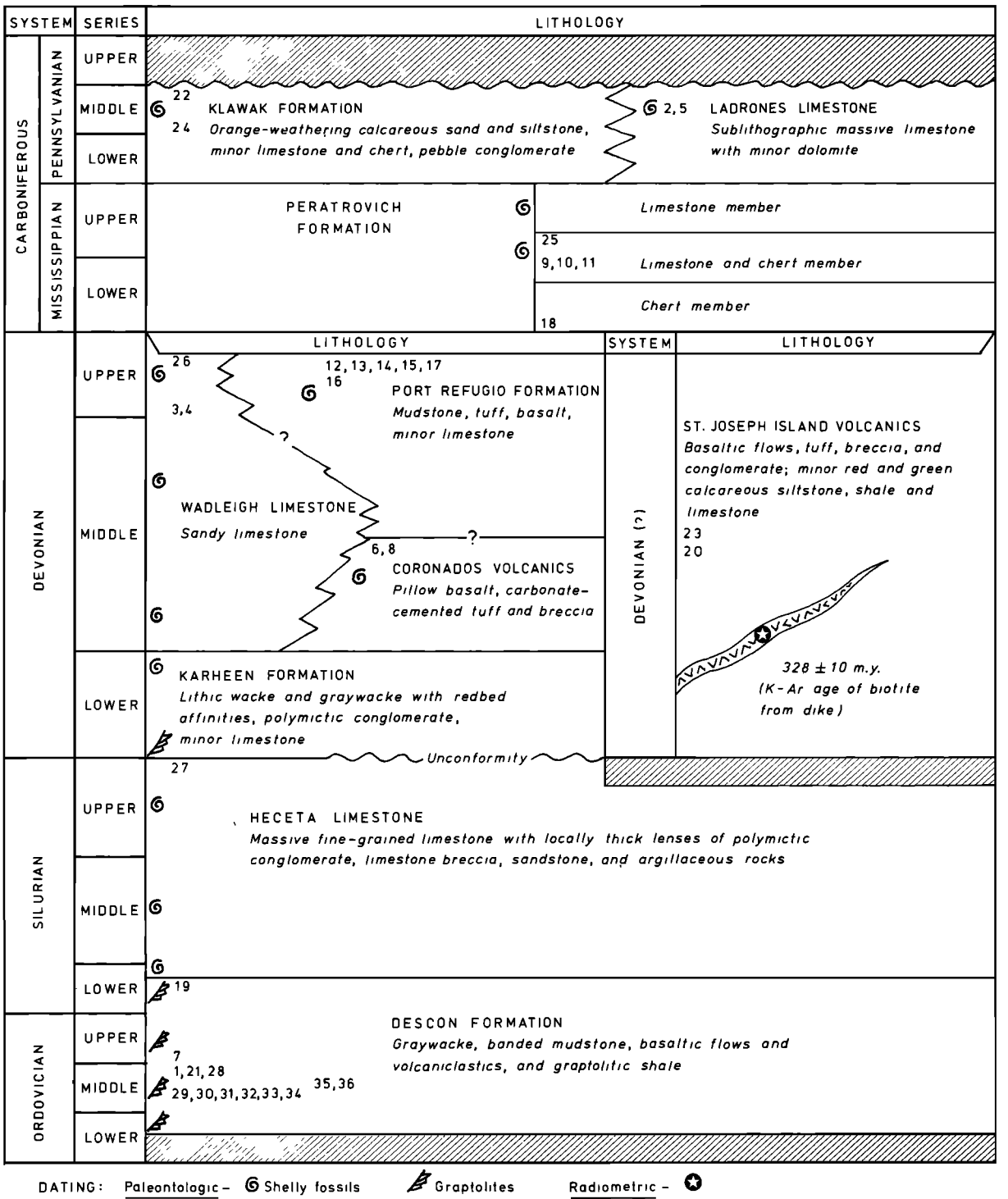


Fig. 2. Schematic stratigraphic column of sampled units in Alexander terrane, after *Eberlein and Churkin* [1970]. Stratigraphic positions of sampling sites are indicated by site numbers.

study area could be discordant with respect to the poles for stable North America. Only geologic and geometric constraints can be used to distinguish the paleomagnetic record of the first hypothesis from those of the others.

To test these hypotheses, paleomagnetic samples were obtained with a portable core drill in and near Prince of Wales Island (Figure 1). The ages of the samples are Middle Ordovician to Middle Pennsylvanian, and the rock types range from submarine and subaerial volcanic rocks and volcaniclastic sedimentary rocks to mudstone and reef limestone (Figure 2). After briefly summarizing the geology of the Alexander terrane, we discuss the paleomagnetic results and compare them with the data available from the North American craton, i.e., that part of present-day North America which can be considered stable and coherent during the Paleozoic Era. We use recently published paleomagnetic results in nearby areas in Alaska [*Hillhouse,* 1977*a*; *Hillhouse and Gromme,* 1980] from rocks of Triassic age to provide additional constraints on the various models.

GEOLOGIC SETTING OF THE ALEXANDER TERRANE

The part of the Alexander terrane investigated in this report is a sequence of Paleozoic rocks that represent every system from Middle Ordovician to Middle Pennsylvanian and is well

TABLE 1. Summary of Paleomagnetic Measurements

Site[1]	Formation, Lithology	R_s[2]	Decl/Incl Before TC[3]	Decl/Incl After TC[3]	k[4]	α_{95}[4]	% Rev[5]	Range[6]	R_3[7]
2	Ladrones, white limestone	5/5	92.3/−32.1	84.3/−20.8	14	21.5	80	435°–600°	4/5
5	Ladrones, red limestone	16/16	88.3/−11.2	87.5/+1.2	18	8.9	100	350°–600°	16/16
22	Klawak, siliceous mudstone	6/6	117.6/−5.2	130.9/−39.0	6	30.0	100	365°–550°	4/6
24	Klawak, siliceous limestone	6/6	141.1/+19.4	133.8/−27.9	8	25.4	83	380°–550°	4/6
9	Peratrovich, gray limestone	6/6	102.1/−32.0	84.8/−37.6	18	16.4	83	30–80 mT	5/6
10	Peratrovich, gray limestone	6/6	121.0/−41.4	103.5/−61.5	86	7.3	100	350°–455°	6/6
11	Peratrovich, gray limestone	4/4	73.7/−8.9	103.0/−17.2	13	26.3	75	350°–455°	2/4
18	Peratrovich, gray limestone	6/6	83.6/−8.4	99.5/−20.0	4	36.1	83	40–100 mT	5/6
25	Peratrovich, gray limestone	6/6	91.2/+36.6	87.8/+14.7	6	29.1	83	300°–410°	3/6
12	Port Refugio, gray silty limestone	6/6	84.9/+2.6	88.5/−24.4	7	27.0	100	380°–500°	2/6
13	Port Refugio, tuffaceous mudstone	5/6	91.1/−30.7	111.4/50.4	20	17.8	100	400°–550°	4/5
14	Port Refugio, tuffs	6/6	82.9/−19.6	94.4/−43.7	6	30.4	100	450°–550°	1/6
15	Port Refugio, massive basalt	0/5	only present-day field magnetizations						
16	Port Refugio, red mudstone	2/7	107.0/+41.5	105.0/−1.1	...	...	100	630°–660°	1/2
17	Port Refugio, aquagene tuffs	11/11	66.0/−34.1	97.4/−39.6	53	6.3	100	380°–500°	10/11
3	Wadleigh, sandy limestone	4/4	54.9/+66.7	288.7/37.6	29	17.0	0	300°–400°	4/4
4	Wadleigh, sandy limestone	5/7	121.5⅛−51.0	107.2/+8.4	159	6.1	100	300°–475°	5/5
26	Wadleigh, silty limestone	7/7	106.3/−6.2	109.8/−34.8	40	9.6	100	350°–475°	5/5
6	Coronados, pillow lava	5/5	only present-day field magnetizations						
8	Coronados, volcaniclastics	5/5	22.2/+67.2	277.8/28.8	384	3.9	0	215°–450°	5/5
20	St. Joseph, red mudstones	10/11	113.3/−51.7	133.8/−9.3	7	19.7	100	620°–675°	3/10
23	St. Joseph, massive basalt and tuffs	0/3	only present-day field magnetizations						
27	Heceta, limestone	2/2	289.7/+38.6	284.6/17.8	...	...	0	165–450°	2/2
21	Descon, banded mudstone and graywacke	0/4	only present-day field magnetizations						
1	Descon, banded mudstone	6/7	120.5/+29.5	118.9/−9.6	6	29.0	100	500°–570°	2/6
7	Descon, quartzo-feldspathic sandstone	3/4	217.1/+68.6	291.0/+15.2	4	76.0	33	450°–550°	2/3
19	Descon, siliceous shale and volcaniclastics	5/7	110.1/−42.7	114.9/−16.9	10	25.1	40	300°–435°	4/5
28	Descon, lava	14/14	80.2/+52.0	83.5/+46.7	34	7.0	100	10 mT	0/14
29	Descon, lava	5/11	42.8/−31.1	100.7/−61.5	29	14.3	100	40 mT	0/5
30	Descon, lava	0/12	only aberrant directions						
31	Descon, lava	9/11	29.0/−39.5	83.2/−71.3	14	14.0	100	30 mT	0/9
32	Descon, lava	10/12	64.8/+4.6	69.2/−16.9	45	7.3	100	40 mT	0/10
33	Descon, lava	7/10	53.5/−3.3	49.2/+15.2	20	13.8	100	40 mT	0/7
34	Descon, lava	11/11	72.4/−36.7	82.7/−2.8	6	19.5	100	40 mT	0/11
35	Descon, lava	15/15	24.3/+67.9	82.2/+2.6	5	20.1	100	20 mT	0/15
36	Descon, lava	14/14	79.7/+24.1	79.5/+59.1	3	27.9	100	20 mT	0/14

[1]Sites correspond to those of Figure 1.
[2]R_s is the ratio of the number of samples used in the calculation of the mean to the number of samples investigated.
[3]Decl/Incl are declination and inclination in degrees; TC is the correction for the tilt of the strata.
[4]k and α_{95} are the statistical parameters of *Fisher* [1953].
[5]% Rev gives the percentage of the number of samples used in the statistical analysis which are interpreted to be of reversed polarity.
[6]Range gives the range of temperature (in °C) or alternating fields in milliTesla (mT) used, in general, to determine the characteristic directions of magnetization.
[7]R_3 is the ratio of samples used in the statistical analysis for which the characteristic directions could be determined from three or more points on the vector diagrams of *Zijderveld* [1967] to the total number of acceptable samples for the site. The origin of the diagram has been included as a point in the case of stable endpoint directions.

exposed along the west coast of Prince of Wales Island and on the adjacent islands to the west. The stratigraphy of these rocks has been described by *Eberlein and Churkin* [1970], who also named the units sampled (Figure 2). The aggregate thickness of the marine sedimentary and volcanic rocks exceeds 11 km. Because most units are richly fossiliferous, they are relatively well dated, and a few radiometric dates supplement the fossil ages.

The base of the oldest unit sampled, the Descon Formation of Early Ordovician to Early Silurian age, is not exposed in the study area, but similar unmetamorphosed Ordovician strata in Klakas Inlet 90 km farther south appear to lie unconformably on the Wales Group, a metamorphosed volcanic-arc assemblage. On southern Prince of Wales Island the Wales Group is cut by an intrusive complex, a trondhjemitic phase of which has yielded discordant zircon populations whose $^{207}Pb/^{206}Pb$ ages fall around 730 m.y. and whose U/Pb ages range from 497 m.y. to 406 m.y. (J. Saleeby, written communication, 1976 and 1979). Granitic and dioritic stocks, several of which have been dated radiometrically, intrude the Paleozoic sequence and reflect episodes of Paleozoic, Mesozoic, and possible early Tertiary plutonism. Sampling near these stocks has been avoided. Apart from the younger intrusive rocks and from one isolated occurrence of Tertiary or Quaternary basalt, the study area is underlain entirely by Paleozoic rocks. The youngest Paleozoic units are the Lower and Middle Pennsylvanian Ladrones Limestone and the Klawak Formation (Figure 2). In addition to their faunal contents, most of the units

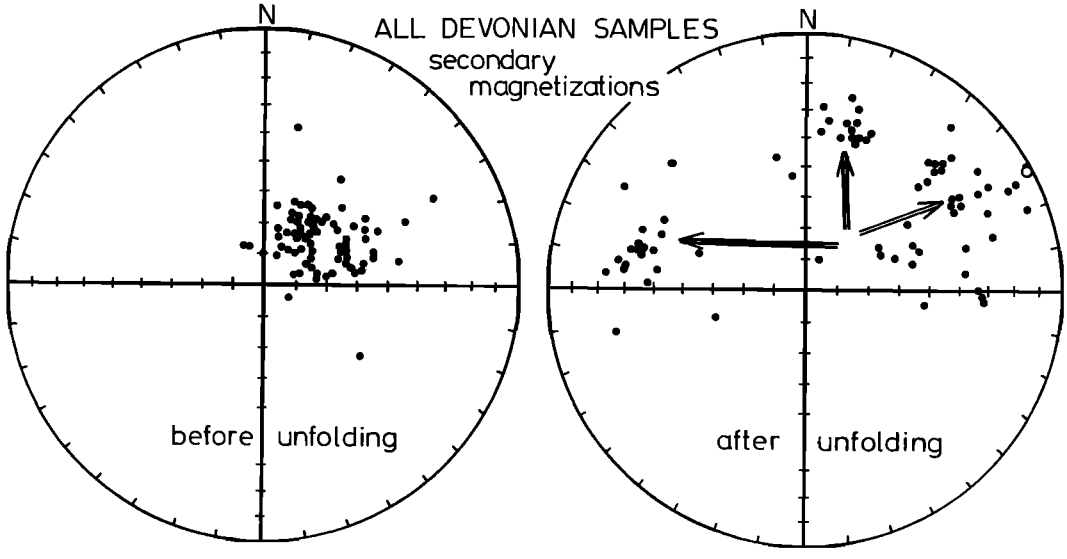


Fig. 3. Equal-area projections of directions of secondary magnetization observed in Devonian samples before and after correction for tilt of strata, respectively. Open circles indicate projections onto upper hemisphere, and solid dots onto lower hemisphere.

sampled show sufficient stratigraphic relationships to establish their relative ages. The only exceptions are sedimentary and volcanic rocks of the St. Joseph Island Volcanics, which occur in an isolated outcrop area in the western islands near the Pacific Ocean (sampled at sites 20 and 23, Figure 1). The St. Joseph Island Volcanics is the only unfossiliferous unit sampled, but its minimum age is established by a K/Ar date of 328 ± 10 m.y. on biotite from a dike that cuts the unit [*Eberlein and Churkin*, 1970]. In addition, on the basis of its lithologic similarities to parts of the Devonian Coronados Volcanics and the Port Refugio Formation, *Eberlein and Churkin* [1970] suggested that the St. Joseph Island Volcanics is best considered to be of Devonian(?) age.

Considering the entire Paleozoic section, the Alexander terrane, with its wide variety of volcanic and plutonic rock types in association with deep- to shallow-water marine sedimentary rocks, has the features of an oceanic floor–island arc setting not unlike those of the underlying Wales Group. Paleozoic geologic history of the terrane begins with the deposition of a thick succession of Lower Ordovician to Lower Silurian

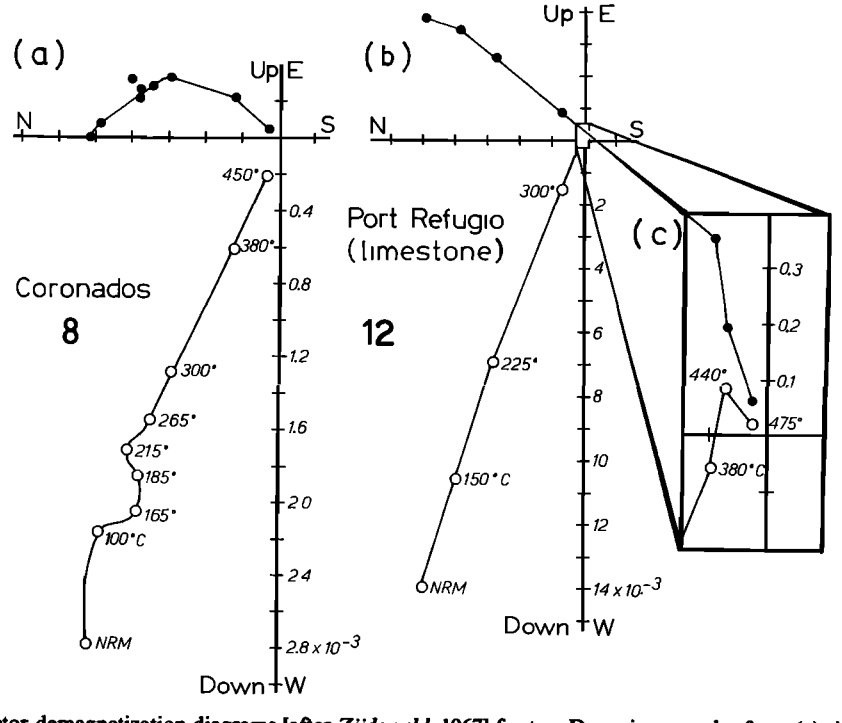


Fig. 4. Vector-demagnetization diagrams [after *Zijderveld*, 1967] for two Devonian samples from (*a*) site 8 and (*b* and *c*) site 12. Successive endpoints of magnetization vector during progressive thermal demagnetization are plotted in in situ coordinates. Solid dots represent projections onto horizontal plane, and open circles onto north-south vertical plane. Units indicated for the axes are in A m^{-1} (10^{-3} emu/cc). Note that enlargement (Figure 4*c*) illustrates high-temperature part of Figure 4*b*.

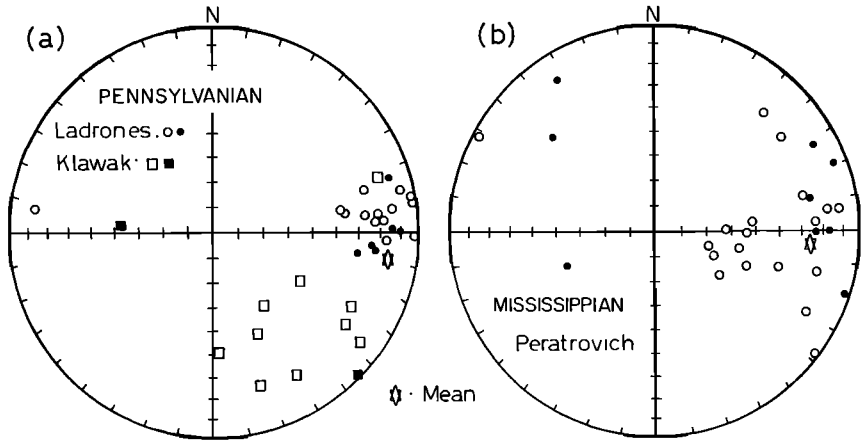


Fig. 5. Equal-area projection of directions of characteristic magnetization from Carboniferous units, corrected for tilt of strata. Circles are for Ladrones, squares for Klawak, and stars for mean directions: Open symbols are on upper hemisphere.

turbidites, interbedded with mafic pyroclastic and epiclastic rocks, pillow basalt, hemipelagic graptolitic shale, and subordinate pelagic chert (Descon Formation) (Figure 2). These strata are overlain by thick, mainly biogenic Lower to Upper Silurian limestone with locally thick lenses and pods of polymictic conglomerate, limestone breccia, and argillaceous rocks (Heceta Limestone), followed by Lower Devonian conglomerate and mudstone (Karheen Formation) having red bed affinities and local depositional features indicative of an intertidal origin [*Ovenshine*, 1975]. Continued upward shoaling is evidenced by an overlying Middle and Upper Devonian coral-reef and reef-related limestone unit (Wadleigh Limestone) that intertongues with vesicular pillow basalt, carbonate-cemented aquagene tuff, and fragmented pillow breccia considered to have been broken up in the surf zone (Coronados Volcanics). Similar volcanic rocks, especially aquagene tuff and breccia, are in the St. Joseph Island Volcanics (Devonian?) and to a lesser extent in the Upper Devonian Port Refugio Formation, which is rich in volcanic material.

After the Devonian, a sequence of chert and chert-bearing limestone (the Mississippian Peratrovich Formation) was deposited in an environment that progressed upward from euxinic, through normal-marine, to clear-water shoaling conditions [*Armstrong*, 1970]. Shallow-water marine conditions prevailed into Middle Pennsylvanian time during deposition of massive limestone and calcarenite (the Ladrones Limestone) and calcareous siltstone, sandstone, and arenaceous limestone with occasional interbeds of fragmented fossil limestone (the Klawak Formation).

The Paleozoic rocks described above have been subjected to at least two major periods of regional folding that formed large-scale open structures whose axes generally trend north to northwest and east to northeast. These structures are transected by several major and many minor faults, some of which can be shown to be of post-Middle Pennsylvanian age, but for the most part their age and history are uncertain because of the absence of younger strata. Northwest-trending faults are the most prominent and, together with the northwest-trending folds, are largely responsible for the dominant northwestward structural grain that characterizes this area and the entire Alexander Archipelago. In many places the latest movement on the northwest-trending fault system can be demonstrated to be overall right-lateral displacement.

PALEOMAGNETIC SAMPLING AND LABORATORY TECHNIQUES

Two of us (C.S.G. and G.D.E.) obtained the first collection of samples from Ordovician lavas of the Descon Formation in 1966. The other samples, from the sedimentary and volcanic units shown in Figure 2, were collected in 1975 by R.V. with help from M.C. and G.D.E. All samples were drilled in the field and oriented (because of overcast skies) with a Brunton compass. The magnetic intensities are too low to have affected the orientation measurements, but because of the relatively high geomagnetic latitude, orientation errors could be as large as ±5°. Because of accessibility and outcrop availability, all sampling sites were located at the shore near or below the high-tide line.

A total of 278 samples were collected at 36 sites (Figure 1) from the units shown in Figure 2. Magnetization of the samples was measured with either a Schonstedt SSM-1A Spinner Magnetometer or the Superconducting Technology cryogenic magnetometers at the University of Michigan and at the U.S. Geological Survey laboratory in Menlo Park, California. (Any use of trade names in this report is for descriptive purposes only and does not constitute endorsement by the U.S. Geological Survey.) The samples with initial intensities below 10^{-3} A m^{-1} (10^{-6} emu/cm^3) were measured with the cryogenic magnetometers. All samples were demagnetized in stepwise fashion using thermal and alternating-field techniques. Demagnetization curves were plotted by the *Zijderveld* [1967] method, and whenever possible (99 out of 136 sedimentary samples; Table 1), directions of magnetization were determined by vector subtraction of straight-line demagnetization trajectories defined by three or more points.

In general, we found that the samples contained two or three vectorially different components of magnetization. In the samples with three components, the component that was first removed had always very low coercivities or blocking temperatures, no more than 3 mT or 100°C, respectively. These magnetizations are considered to be of viscous origin, acquired during storage in the laboratory. Of the other two components of magnetization, those with the lower coercivities or the lower blocking temperatures, had directions that agreed closely with the present-day magnetic field direction in southeastern Alaska. These components were found in most

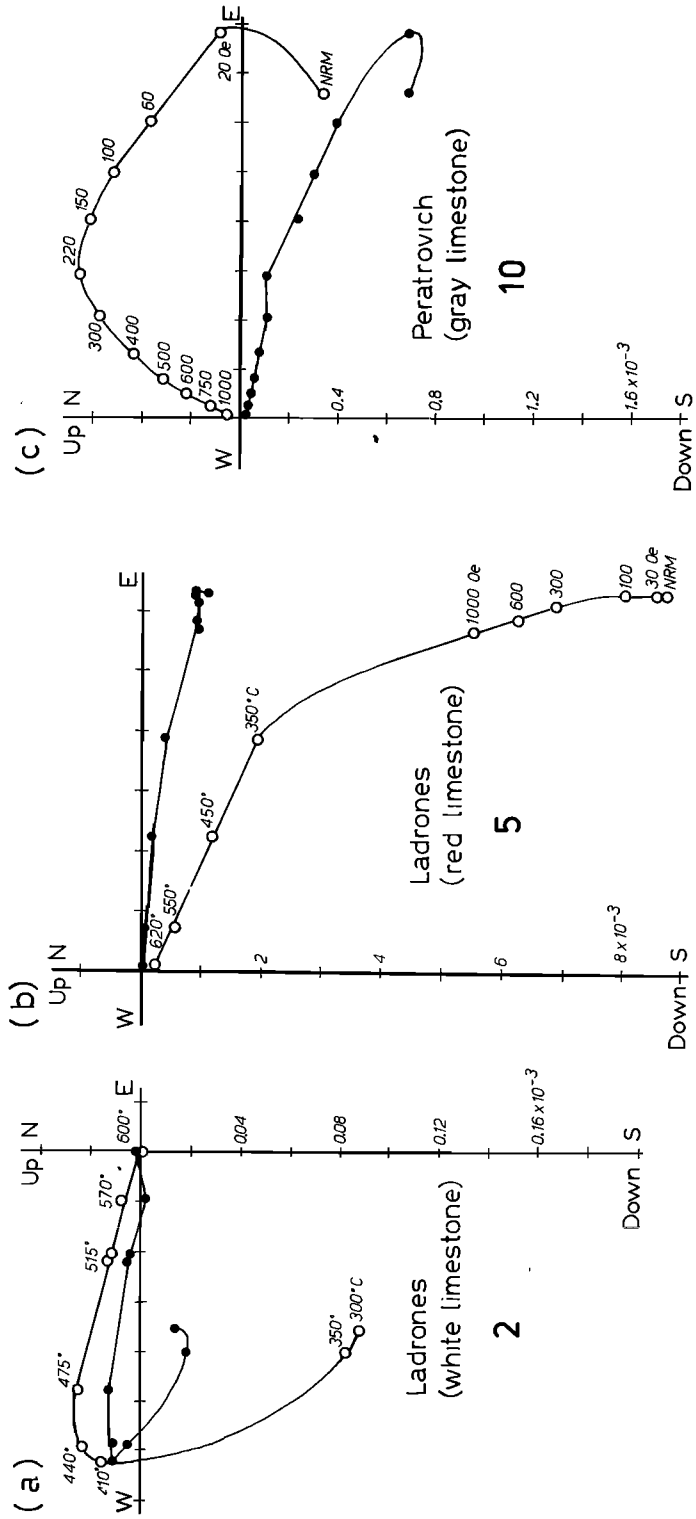


Fig. 6. Vector-demagnetization diagrams for three Carboniferous samples, plotted after correction for tilt of strata. For clarity, vertical projections are here plotted for east-west vertical plane. Samples were from (*a*) site 2, (*b*) site 5, and (*c*) site 10, respectively. Symbols same as in Figure 4.

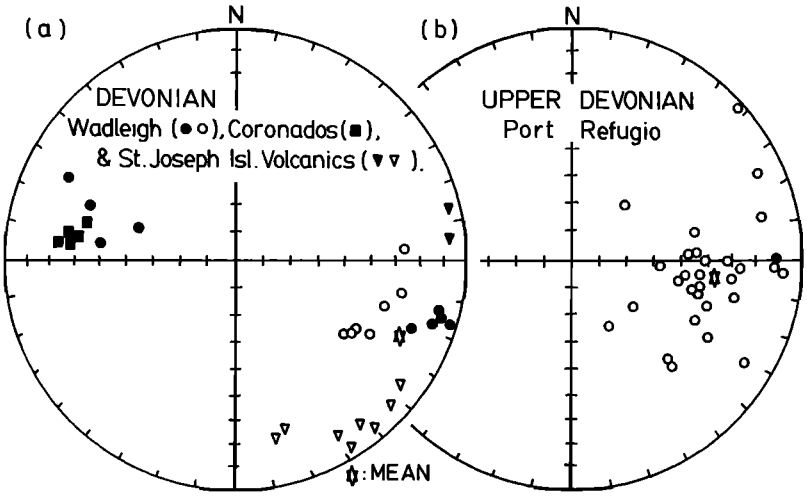


Fig. 7. Equal-area projections of directions of characteristic magnetization from Devonian units, corrected for tilt of strata. Circles are for Wadleigh, squares for Coronados, and triangles for St. Joseph Island. Open symbols are on upper hemisphere.

of the samples. As an example, Figure 3 shows these directions for all Devonian samples, before and after correction for the tilt of the strata. Because the directions disperse greatly upon unfolding, we conclude that these magnetizations are secondary and most likely of fairly recent origin. In 12.2% of all samples, only ths present-day-field (pdf) direction was found; these samples, including all samples from sites 6, 15, 21, and 23, will not be discussed further.

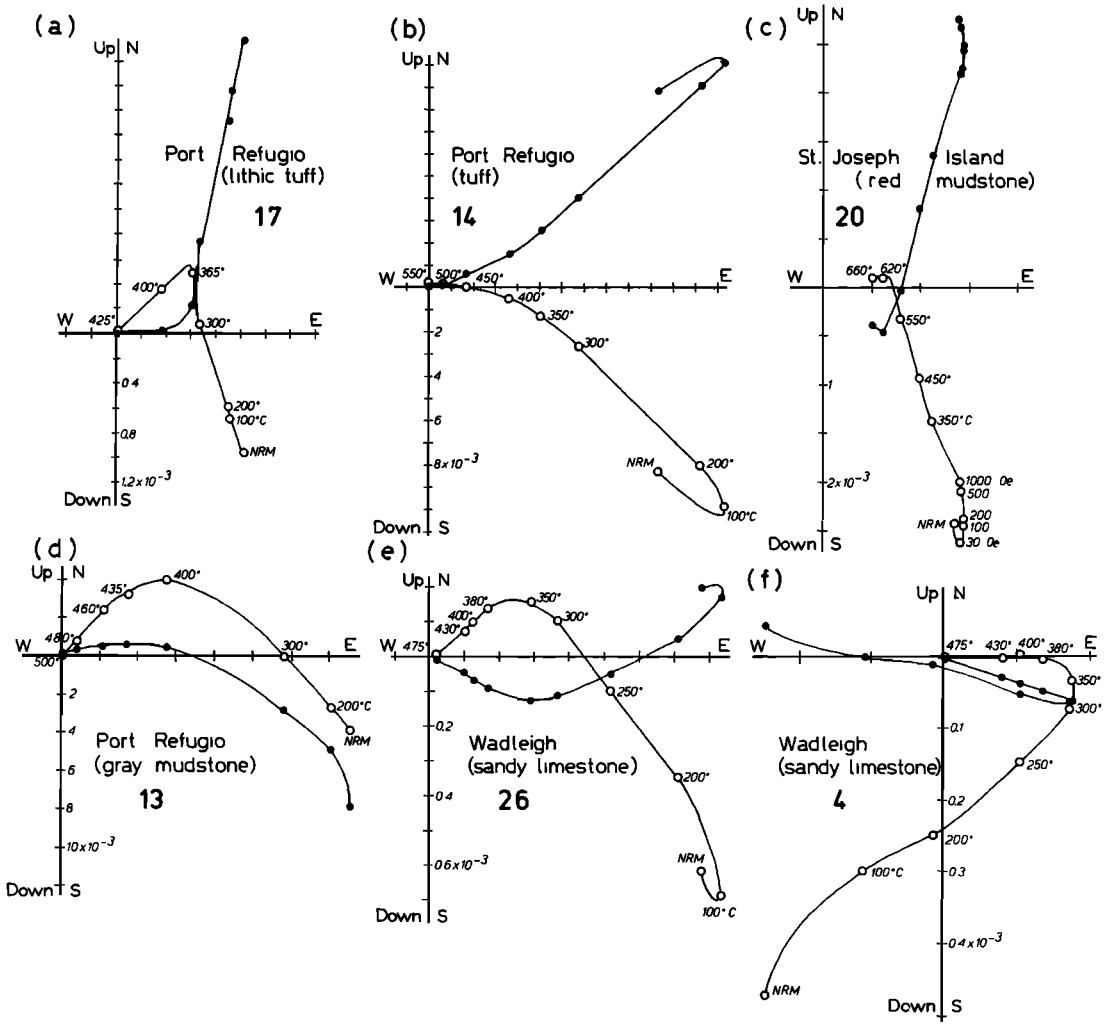


Fig. 8. Vector-demagnetization diagrams for six Devonian samples, plotted after correction for tilt of strata. Samples were from sites (*a*) 17, (*b*) 14, (*c*) 20, (*d*) 13, (*e*) 26, and (*f*) 4. Symbols same as in Figures 4 and 6.

TABLE 2. Formation Means and Pole Positions

Age	Formations	R_{sa}	R_{si}	k_1	$(\alpha_{95})_1$	Decl/Incl, deg	k_2	$(\alpha_{95})_2$	k_2/k_1	Pole Position	dp	dm	p	Δ_{95}
Cu	Ladrones Ls.	33/33		5.9	11.3	100.5/−15.1	5.6	11.6	0.95	12.3°N, 132.6°E	6.1	11.9	82.3	12.0
	and Klawak Fm.		4/4	6.6	38.7	107.3/+23.4	7.5	35.9	1.14					
Cl	Peratrovich Fm.	28/28		4.3	14.9	94.6/−26.5	5.1	13.3	1.19	14.1°N, 141.2°E	7.8	14.4	76.0	14.9
			5/5	5.8	34.9	95.2/−24.6	8.0	28.7	1.38					
Du	Port Refugio Fm.	30/41		6.6	11.1	97.5/−37.1	10.1	8.7	1.53	21.1°N, 142.9°E	6.0	10.2	69.3	10.9
			5/6	6.0	34.2	98.9/−32.3	15.2	20.3	2.53					
D	Wadleigh Ls.,	31/42		4.6	13.5	114.9/−19.9	8.6	9.4	1.87	22.5°N, 121.9°E	5.2	9.8	79.7	10.0
	Coronados and		5/7	4.5	40.6	111.9/−21.0	12.3	22.7	2.73					
	St. Joseph													
	Island Volc.													
Ou	Descon Fm., Sedimentary	14/22		2.4	33.2	115.5/−13.5	7.4	15.7	3.08	19.9°N,118.9°E	8.2	16.0	83.2	16.2
	rocks		3/4	2.1	>90.0	115.0/−13.9	224.0	8.3	107.40					
Om-u	Descon Fm., lavas		8/9	4.0	34.6	76.4/−2.7	3.0	37.8	0.75	4.0°S,148.8°E	18.9	37.8	88.6	37.8

Explanation: Age: Om-u = Middle and Late Ordovician, Ou = Late Ordovician, D = Devonian, Du = Late Devonian, Cl = early Carboniferous, Cu = late Carboniferous; R_{sa} is ratio of number of samples used for calculation of mean to number of samples investigated; R_{si} is a similar ratio for number of sites; k and α_{95} are precision parameter and semiangle of cone of confidence at 95% probability level, respectively [*Fisher*, 1953], with subscript 1 denoting these values before and subscript 2 the values after correction for tilt of strata; Decl/Incl are declination and inclination after correction for tilt of strata; dp and dm are semiminor and semimajor axes, respectively, of oval of 95% confidence around pole; p is distance to paleopole; and Δ_{95} is uncertainty associated with declination at 95% probability level ($\Delta_{95} = \sin^{-1}$ [Sin α_{95} / cos (Incl)]).

The data from 3 of the 36 sites (3, 8, and 35, Figure 1) are somewhat ambiguous in that before correcting for the tilt of the strata the magnetization directions are close to the pdf direction, but after making the corrections the directions are similar to those for other sites in the same unit. We have decided to include the results from these sites in the final calculations because the samples usually showed good demagnetization behavior (e.g., Figure 4*a*), with straight-line decay to the origin of the vector diagram, and did not show evidence of other high-coercivity or high-blocking-temperature magnetization directions (such as those shown at ≥380°C and higher temperatures in Figure 4*c*.

PALEOMAGNETIC RESULTS

Carboniferous units. Samples from the two Pennsylvanian units, the Ladrones Limestone and the Klawak Formation, as well as the Lower Mississippian part of the Peratrovich Formation, showed directions after demagnetization that grouped around east to southeast declinations, with mostly negative (upward) inclinations (Figure 5). Mean directions for the sample sites are listed in Table 1; no samples of these units had to be rejected. For the Klawak and the Peratrovich the fold test was positive, although this result is not significant for the Peratrovich. For the Ladrones Limestone the two sites had the same tilt correction.

Demagnetization diagrams for the Carboniferous units are shown in Figure 6. Depending upon the rock type involved, the coercivities or blocking temperatures of the characteristic magnetizations were intermediate (for the calcareous sandstone of the Klawak and the limestone of the Peratrovich) to high (for the red and white limestones of the Ladrones). Fairly large secondary magnetizations with a pdf direction were removed before the characteristic magnetizations were revealed. Normal and reversed polarities are present in samples from all three units (Figure 5; compare also Figures 6*a* and 6*b*), but the majority of samples had a polarity that we interpret as reversed. Although the number of normal-polarity directions is too small for a statistical treatment to determine the consistency of the reversals, the approximately antipodal nature of the opposing polarities suggests that little if any secondary overprinting contaminates the directions of magnetization plotted in Figure 5.

The directions are grouped by age, as Pennsylvanian or Mississippian, and the associated cones of 95% confidence (α_{95}) are 11.6° and 13.3°, respectively, with unit weight given to samples in the statistical analysis. Because the sampling was carried out in order to maximize stratigraphic coverage in a given site (up to 40 m) and because the units sampled are sedimentary rocks, which presumably average out the Carboniferous secular variation if enough stratigraphic thickness is sampled, this appears to us to be the preferable weighting procedure. For comparison, however, the mean directions obtained by giving unit weight to the sampling sites are also listed (Table 2). Because of the relatively smaller number of sites used in the latter calculation, these directions have correspondingly larger α_{95}, although they are approximately the same.

A difference, most pronounced in the declinations, exists between the results from the Ladrones Limestone and those from the majority of samples from the Klawak Formation. These units are probably of the same age, although they may possibly have been rotated relative to each other. Supracrustal-block rotations are indeed conceivable for all areas of the Alexander terrane, and *Eberlein and Churkin* [1980]

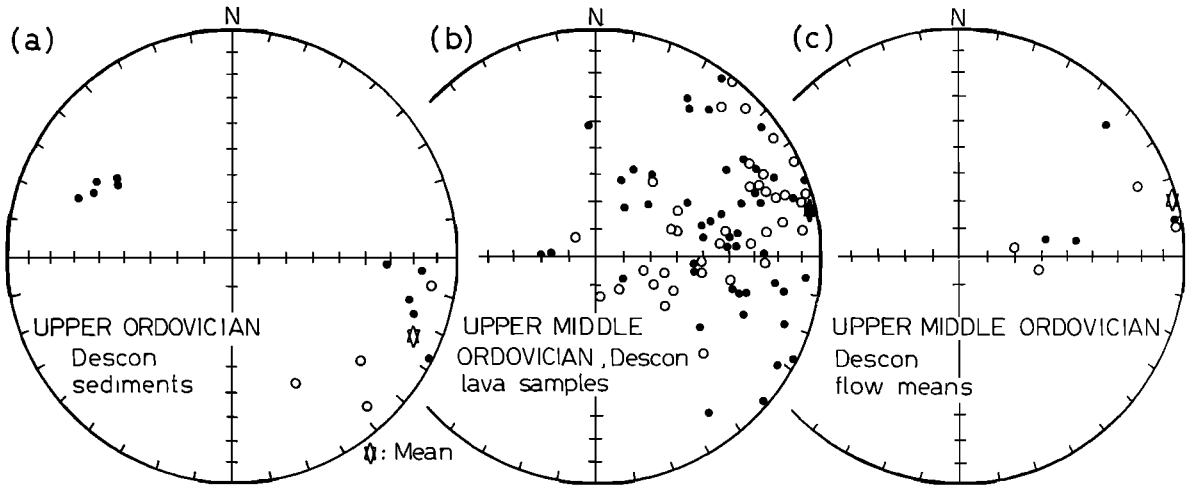


Fig. 9. Equal-area projections of directions of magnetization from Ordovician sedimentary and volcanic rocks of the Descon Formation, after demagnetization and corrected for tilt of strata. Each point in Figures 9*a* and 9*b* represents one sample; Figure 9*c* shows plot of site means. Symbols same as in Figures 3 and 5.

mapped several faults that can accommodate such rotations. On the other hand, we sampled the Mississippian Peratrovich Formation near outcrops of both the Klawak and the Ladrones, and yet its paleomagnetic declinations do not indicate relative rotation of the two areas. Alternatively, plunging fold axes in the structurally complex area of the Klawak Formation may be responsible because they affect the declinations when tilt corrections are made by simple rotation about the strike. We defer further discussion of this problem to a later section, in which we compare the paleomagnetic results with each other and with those from the North American craton. Our major tectonic conclusions are based on the inclination data, which are the same for both the Ladrones Limestone and the Klawak Formation.

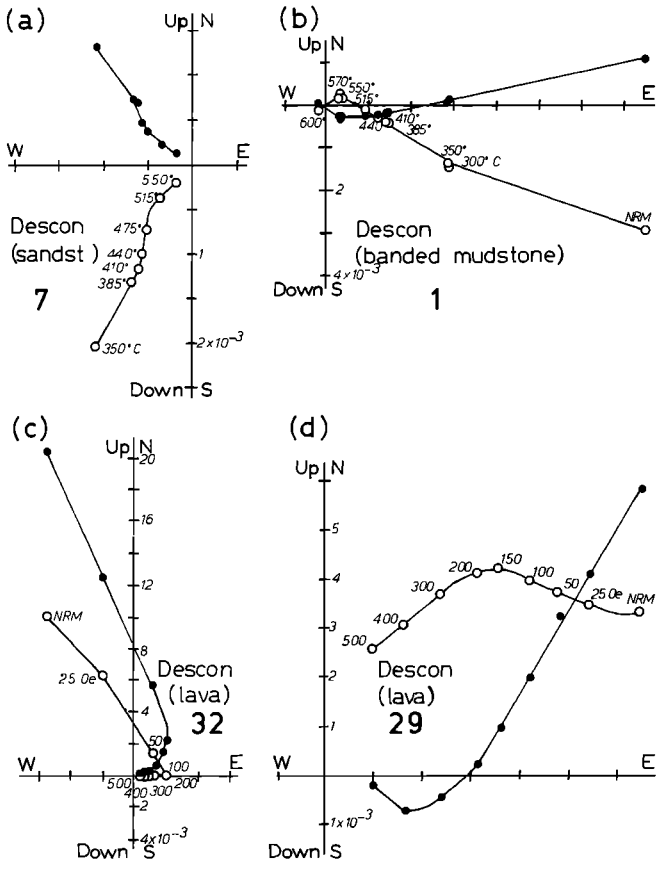


Fig. 10. Vector-demagnetization diagrams for (*a* and *b*) two sedimentary rock samples and (*c* and *d*) two lava samples of the Descon Formation, plotted after correction for tilt of strata. Samples were from sites (*a*) 7, (*b*) 1, (*c*) 32, and (*d*) 29. Symbols same as in Figures 4 and 6.

Devonian units. The Devonian samples are divided into two groups: group 1 includes the well-dated Upper Devonian Port Refugio Formation; group 2 includes the richly fossiliferous Middle and Upper Devonian Wadleigh Limestone, the Middle Devonian Coronados Volcanics, and red mudstones of the unfossiliferous Devonian(?) St. Joseph Island Volcanics (see also Figure 2). The Karheen Formation also was sampled but will not be discussed further because its coarse-grained sandstone showed unstable and uninterpretable magnetic behavior during pilot thermal or alternating-field demagnetization.

After removal of the lower coercivity or lower blocking temperature (pdf) components, the directions in the other Devonian samples (Figure 7) show eastward to southeastward declinations and mostly negative (upward) inclinations, as do the Carboniferous rocks. With unit weight given to samples, the fold tests for the two groups are statistically significant and positive (Table 2). The mean reversed-polarity direction in Figure 7*a* (D = 119.9°, I = −13.8°, N = 22, k = 7.6, α_{95} = 12.1°) is 160° away from the normal-polarity mean direction (D = 282.3°, I = +32.8°, N = 9, K = 47, α_{95} = 7.6°). The fact that these two mean directions are not exactly antipodal is due largely to the low (positive) inclinations in site 4 in the Wadleigh Limestone (Table 1). Secondary magnetizations may still contaminate the directions at this site, although the demagnetization diagrams (e.g., Figure 8*f*) show no evidence of this effect.

Figure 8, as well as Figures 4*b* and 4*c*, presents demagnetization diagrams showing that, again, the secondary magnetizations of low blocking temperature are large. However, most of the characteristic magnetization directions could be fairly accurately determined, even when they constituted less than 3% of the total natural remanent magnetization (NRM), as shown for example in Figures 4*b* and 4*c*. In most rock types, such as the gray mudstone, tuff, and sandy limestone, the secondary magnetizations are eliminated at about 400°–430° C, whereupon the remaining magnetization decays to the origin.

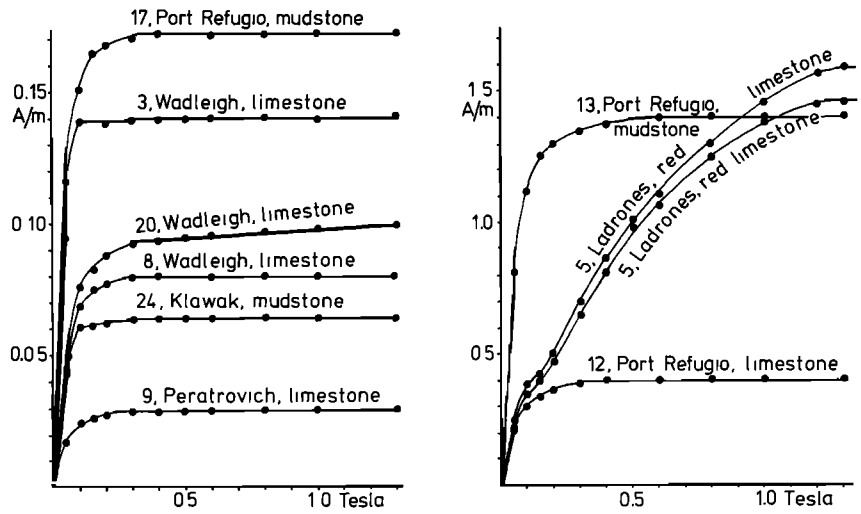


Fig. 11. Diagrams showing acquisition of isothermal remanent magnetization (IRM) in several Devonian and Carboniferous limestone samples (after *Dunlop's* [1972] method).

Only the red mudstones of the Port Refugio Formation (two samples, site 16, Figure 7) and the St. Joseph Island Volcanics (site 20, Talbe 1; Figure 8*c*) show much higher blocking temperatures and coercivities due to the predominance of hematite as the magnetic carrier.

The group means have associated α_{95} of 8.7° and 9.4° for the Upper Devonian Port Refugio Formation and the combined other Devonian units, respectively (Table 2). Site 20 has somewhat higher (more southeastward) declination values but agrees in inclination. Again, block rotations or plunging fold axes may be responsible.

Silurian units. Owing to a breakdown of the drilling apparatus, only two samples of the Silurian Heceta Limestone could be obtained at one site (27, Figure 1). Although the results from these samples (Table 1) have no statistical significance, they are mentioned here because they show directions similar to those in the underlying Descon Formation.

Ordovician and Early Silurian units. Both sedimentary rocks (mainly volcaniclastic graywacke and mudstone) and basaltic lava flows of the Descon Formation were sampled. The Descon, whose base is not exposed, is known to range in age from at least Arenigian (late Early Ordovician) through late Llandoverian (Early Silurian). The sedimentary rocks samples for this report range in age from Middle Ordovician (early Caradocian) to earliest Silurian (middle Llandoverian). The basaltic flows are all considered Middle Ordovician (Caradocian).

The Descon is essentially unmetamorphosed. The graywacke is composed mainly of relatively fresh mineral and intraformational lithic fragments set in a chloritic matrix that can constitute as much as 25% of the rock. Diagenetic analcime and leonhardite and a minor amount of carbonate may also be present as a cement. Mudstone, interbedded with the graywacke in thin-bedded sets generally less than 2.5 cm thick, typically has a banded appearance. The basaltic lavas are predominantly porphyritic with stubby phenocrysts of diopsidic augite and rare plagioclase set in a groundmass of smaller granular augite, plagioclase microlites, and chloritic alteration products probably derived from material that was originally glass. The pyroxene is unaltered, but the plagioclase usually shows some degree of alteration to calcite, clinozoisite, prehnite, and various zeolites [*Eberlein and Churkin*, 1970].

Two-thirds of the sedimentary rock samples yielded high-temperature magnetizations (150° to 570°C) that group around a west-northwest, east-southeast declination axis with shallow inclinations; both normal and reversed polarities are observed (Figure 9*a*). The fold test on these data, giving unit weight to either samples or sites, is positive and statistically significant at the 99% confidence level (Table 2). Demagnetization diagrams for a normal- and a reversed-polarity sample are shown in Figures 10*a* and 10*b*. The characteristic magnetizations were generally revealed above 500°C or above 30 mT (300 Oe), and demagnetization was complete by 570°C or 100 mT. The mean reversed-polarity direction (D = 114.0°, I = +0.1°, N = 9, k = 7, α_{95} = 21°) is 145° away from the normal-polarity mean direction (D = 298.4°, I = +35.1°, N = 5, k = 163, α_{95} = 6.0°). The scatter is relatively large in the reversed-polarity directions of the high-temperature magnetizations (e.g., Figures 9*a* and 10*b*). Secondary magnetizations may still contaminate to some extent the characteristic directions chosen. From 14 samples (3 sites), a mean direction was obtained with D = 115.5°, I = −13.5°, and α_{95} = 15.7° (Table 2). We note that the mean direction of the Heceta site (D = 284.6°, I = +17.8°) lends support to the direction in the Descon sedimentary rocks.

Nine lava flows were sampled, and 110 individually oriented cores were obtained. Through alternating-field demagnetization up to 40 mT or higher, we obtained directions of magnetization that generally appeared to decay toward the origin (Figure 10*c*). These directions could not be accurately determined from the Zijderveld diagram in all cases because at fields of 40 to 50 mT, significant spurious magnetizations commonly were introduced by tumbling in the alternating field demagnetization apparatus. All demagnetization steps were done twice to average these effects vectorially [*Hillhouse*, 1977*b*], and directions were selected before the samples showed large vectorial differences between the two measurements. The best step for each site was judged from the Zijderveld diagrams and the movement of the mean directions.

Seven samples from various sites that showed directions widely divergent from their site means at the demagnetization level chosen have not been included in the final calculations. Similarly, directions in one entire sampling site (30, Figure 1) were far removed from the others and have been discarded. The sedimentary rocks near this site are much more deformed than at the other sites. At site 29, 6 of the 11 samples were re-

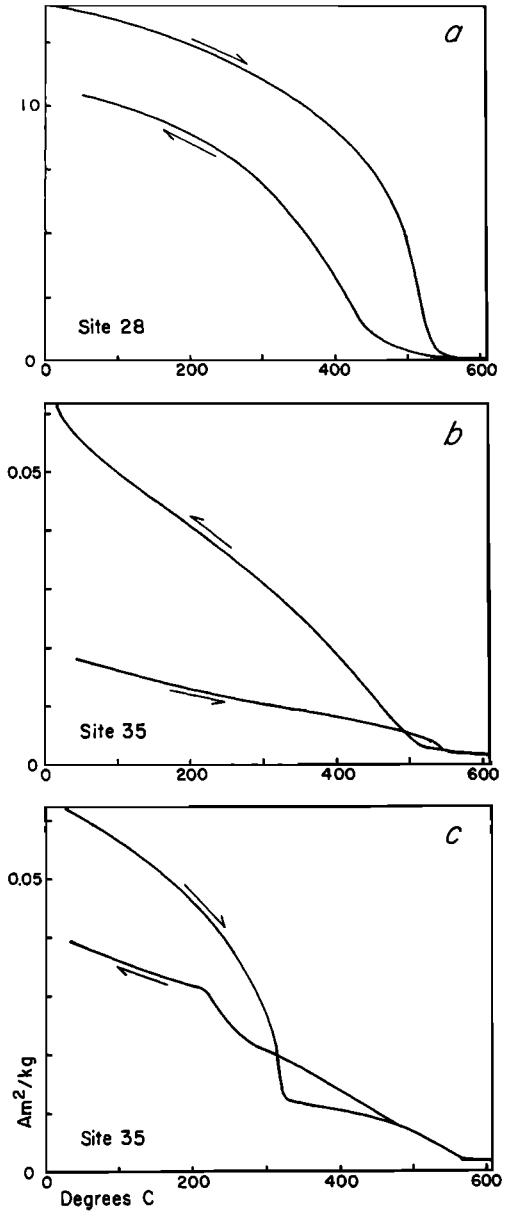


Fig. 12. Strong-field thermomagnetic curves for two Descon lava flows. Experiments were done in a nitrogen atmosphere; magnetizing field was 0.25 T for site 28 and 0.40 T for site 35. Heating and cooling curves are indicated by arrows. (SI units of specific magnetization: 1 Am^2/kg = 1 emu/g.)

jected because they were remagnetized by a Tertiary dike that cuts the outcrop. (At the time of sampling, the age of the dike had not been determined.) The fold test on data from the eight selected sites (or alternatively from all nine sites) is inconclusive because the site mean directions are rather scattered both before and after unfolding, with corresponding α_{95} of 34.6° and 37.8°, respectively. The site mean directions are distributed in a strongly linear fashion, resulting in a large uncertainty in the inclination. By itself, the result from the lavas of the Descon would not lead to any conclusion about the history of the Alexander terrane, but the results from the associated sedimentary rocks of the Descon lend support to their mean direction. The 'failure' of the fold test and the distribution of the site means cannot be due to noncylindroidal folding or plunging fold axes, because these conditions would cause greater uncertainty in the declination. All bedding attitudes were measured on intercalated sedimentary rocks, so that the main reason for the site mean distribution must be sought in secular variation and partly in unremoved secondary magnetizations. We note that if the steep upward directions at sites 29 and 31 represent two components of magnetization, the remaining secondary magnetizations do not lie along the pdf direction (Figure 10*d*).

CARRIERS OF THE MAGNETIZATION

We unsuccessfully attempted to leach a few limestone samples in order to identify magnetic minerals in the residue; only an occasional pyrite grain was observed in addition to the nonferrous minerals. More successful was the technique of imparting an isothermal remanent magnetization (IRM) to the carbonate samples. As discussed by *Dunlop* [1972] and *Heller* [1978], acquisition of IRM can be quite diagnostic because the coercivities of magnetite, hematite, and goethite generally differ significantly. Figure 11 shows IRM acquisition curves for a representative set of samples from the sedimentary units. With the exception of two red limestone samples, all samples reach saturation at or below ≤0.3 T (3000 G), a result indicating titanomagnetite or magnetite as the principal carriers of the magnetization. We assume that these magnetites are detrital or of early bacterial origin and that therefore most of the observed characteristic directions are primary. Red limestone of the Ladrones Limestone and the two sites (16 and 20, Figure 1) of Devonian red mudstone may carry a chemical remanence in hematite, but we see no reason to postulate a secondary origin for their magnetization directions.

We obtained strong-field thermomagnetic curves for one or two samples from each site in the Descon lavas; the range of behavior is shown in Figure 12. Only one of the lavas, site 28, had an NRM intensity comparable to fresh basalt (1.0 A/m). The Curie temperature was 535°C (Figure 12*a*), and heating above 600°C altered the sample so as to reduce both the Curie temperature and the magnetization. This type of behavior is typical of titanium-poor titanomagnetite containing ilmenite lamellae that were formed naturally as the result of high-temperature deuteric oxidation, and hence indicates primary igneous oxide minerals. The decrease in Curie temperature and magnetization are due to resorption of some ilmenite lamellae back into the titanomagnetite. The thermomagnetic curve for site 35 (Figure 12*b*) is typical of 7 of the 10 samples analyzed. Very little magnetic oxide is present, and the Curie temperatures range from 530° to 575°C (547°C in the example shown). The large increase in magnetization after heating to 600°C is apparently due to breakdown of chlorite to magnetite and other materials, which commences at about 580°C. The average NRM intensity of the six sites for which only this type of thermomagnetic curve was obtained is 7.9×10^{-3} A/m, more than 100 times weaker than at site 28. A second thermomagnetic curve for site 35 is shown in Figure 12*c*; the presence in this specimen of pyrrhotite (Curie temperature 325°C) in addition to titanium-poor titanomagnetite is unambiguous [*Schwarz*, 1975], and pyrrhotite also was found in a sample from site 31. The average NRM intensity at sites 31 and 35 is 1.0×10^{-1} A/m and is much more variable than at the other sites, a result in keeping with the apparently nonuniform distribution of pyrrhotite. The fact that the Curie temperatures of titanomagnetite observed in all the lavas is signif-

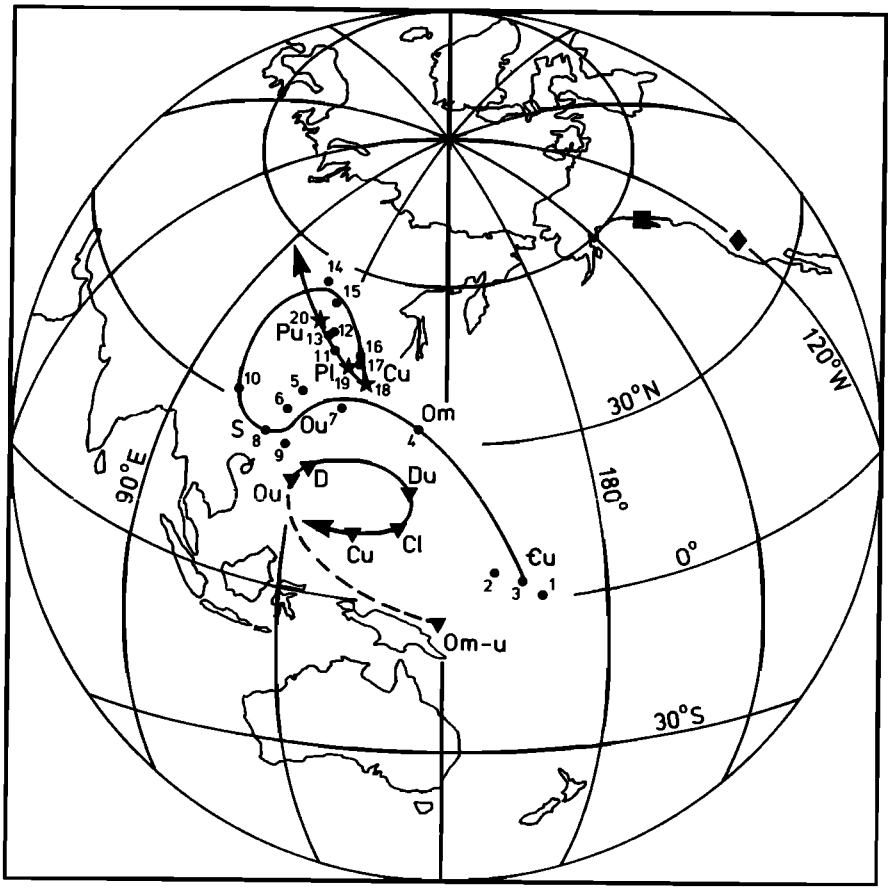


Fig. 13. Plot of pole positions from this study (triangles) and from North American craton (dots). Numbers correspond to references in Table 3. The ovals of 95% confidence (listed in Table 2) have not been plotted for clarity. Further statistical analysis is shown in Figure 14 and discussed in text. Ages: ϵ = Late Cambrian, Om = Middle Ordovician, Omu = Middle and Late Ordovician, Ou = Late Ordovician, S = Silurian, D = Devonian, Du = late Devonian, Cl = early Carboniferous, Cu = late Carboniferous, Pl = early Permian, Pu = late Permian. Square is present location of sampling area in Alexander terrane; diamond is inferred approximate original location.

icantly below that of pure magnetite (585°C) indicates that it is of primary igneous origin. The pyrrhotite is also apparently primary because the NRM in both the pyrrhotite and titanomagnetite have the same directions.

DISCUSSION

We now briefly discuss the paleomagnetic results for the Paleozoic rocks of the Alexander terrane and then compare these data with the available contemporaneous data for the North American craton.

The collection of samples from the Alexander terrane has yielded six group averages for rocks of late Middle Ordovician, Late Ordovician, Devonian, Late Devonian, early Carboniferous, and late Carboniferous ages. These data and the corresponding paleomagnetic pole positions are summarized in Table 2. For the sedimentary units, the pole positions have been calculated from the mean directions, giving unit weight to samples in the statistical analysis. The sedimentary groups generally yielded α_{95} much less than 15°, resulting in acceptable ovals of confidence about the pole defined by semiminor axis *dp* and semimajor axis *dm* (listed in Table 2). Only the late Middle Ordovician lavas of the Descon Formation had a rather large α_{95} of 37.8°, giving unit weight to sites ($N = 8$).

The pole positions have been plotted in Figure 13 (triangles), where it is assumed that these are north poles on the basis that the simplest movement of the least magnitude (discussed below) is preferable. These poles are to be compared with the poles from cratonic North America, also plotted in Figure 13. For clarity, the corresponding ovals of confidence are not plotted, but the values of *dp* and *dm* are listed in Table 2.

As mentioned above, at some sites the sedimentary rocks yielded differences in declination that appear larger than those normally associated with secular variation. Because the Alexander terrane during the Paleozoic must have resembled an island arc and should therefore be classified as a mobile belt, supracrustal rotations could conceivably have occurred within the terrane. Also, we reiterate that with plunging fold axes and/or noncylindroidal folding, the declinations may have been affected.

On the other hand, the inclinations of the NRM vectors referred to the bedding planes are not affected by this type of deformation, and we have no reason to doubt their validity. Normal and reversed polarities were commonly observed, and the demagnetization diagrams generally show enough discreteness in coercivity or blocking temperature to enable separation of secondary (pdf) and characteristic magnetizations. Thus it appears preferable first to address the issues related to

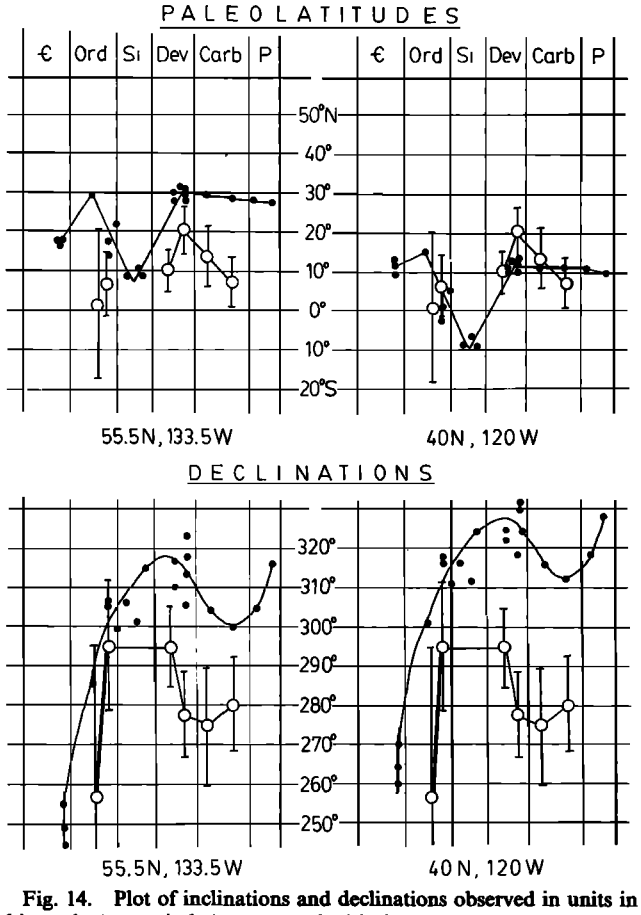


Fig. 14. Plot of inclinations and declinations observed in units in this study (open circles), compared with those predicted (by extrapolation) from North American craton (solid dots). Two positions, discussed in text, have been used to plot these predictors: 55.5°N, 133.5°W (i.e., present position of Alexander terrane) and 40°N, 120°W (most plausible paleoposition of Alexander terrane). Error bars for inclination correspond to *dp* (listed in Table 2), and those for declination are derived from α_{95} (also in Table 2). Explanation of ages: € = Cambrian, Ord = Ordovician, Si = Silurian, Dev = Devonian, Carb = Carboniferous, and P = Permian.

inclination and corresponding paleolatitudes. The paleolatitudes are plotted in Figure 14 (open symbols), where the vertical bars represent the associated errors in paleolatitude given by the *dp* values.

The currently available data from North America for the Paleozoic, listed in Table 3, show a reasonable coverage for all periods, especially the late Carboniferous and the Permian, so that we have listed only their poles, as determined by *Van der Voo and French* [1974]. For the Middle and Late Devonian we refer to a discussion by *Kent and Opdyke* [1978], and we have not included any poles from areas that possibly did not belong to the North American craton, such as eastern New England, New Brunswick, and Newfoundland. It turns out that the Middle and Late Devonian poles from the craton are very close to those for the Late Permian, and it has been argued by some workers that the Devonian rocks possibly were remagnetized during Late Permian time. However, earlier results for the folded Catskill Redbeds in Pennsylvania [*French*, 1976] have recently been confirmed by a conclusive fold test [*Van der Voo et al.*, 1979], and the Late Devonian Catskill poles (11 and 12, Figure 13 and Table 3) must be considered older than Permian. The presence of antipodal reversals in these rocks lends additional support to an older, presumably Late Devonian, age of magnetization. The available Late Cambrian, Ordovician, and Silurian poles were discussed by *Van der Voo and French* [1977] and *Watts and Van der Voo* [1979] and for each period these poles form rather well defined clusters (Figure 13 and Table 3).

From these pole positions we have calculated the paleolatitudes expected for the present-day position (55.5°N, 133.5°W) of the Alexander terrane (dots, Figure 14; Table 3). The observed inclinations have been used to compute the paleolatitudes (circles, Figure 14) of the Alexander terrane, assuming that most directions were reversed and that the poles in Figure 13 are north poles. Had the alternative polarities been chosen, the resulting discrepancies would be much larger. The error bars for the Alexander terrane poles do not overlap with the predicted values (dots), and the paleolatitude values therefore differ significantly from those expected for the terrane's present-day position. Thus, a more southern paleoposition is favored, and the error limits allow us to suggest a Paleozoic position for the Alexander terrane. Influenced by the paleoposition suggested by *Jones et al.* [1972], we have selected 40°N, 120°W (see Figure 1 inset for location) and note that for this paleoposition the predicted and observed paleolatitudes appear to agree optimally (Figure 14).

Also plotted in Figure 14 are the declinations, extrapolated from cratonic North America to the present-day position as well as to the possible paleoposition of 40°N, 120°W. Here again the error bars, functions of the α_{95} associated with the Alexander terrane observations, do not overlap with the predicted values, at least for the Devonian and Carboniferous units. For the present-day position, the individual results show deviations ranging from 10° to 37° when the error bars are ignored, with an average of about 25° in a counterclockwise sense. We interpret this to indicate a post-Carboniferous counterclockwise rotation of the terrane with respect to the present-day grid meridian at 55.5°N, 133.5°W. Despite the individual variations in declinations discussed above—whatever their cause—these declination differences appear to be rather consistent for the Paleozoic data. Therefore, given the wide distribution of sampling sites (Figure 1), we conclude that the entire study area must have moved more or less as a block. For the paleoposition at 40°N, 120°W, the declination differences are even larger and average 35°. The choice between the two values (25° or 35°) depends on whether the rotation (calculated with respect to the paleomeridians) took place before the northward drift began or after it had been completed.

We have also attempted to match the apparent polar-wander path segments of the Alexander terrane and the North American craton (Figure 13), assuming that the two regions did not move relative to one another during the part of Paleozoic time in which Alexander rocks formed. Unfortunately, some of the large ovals of confidence and small rates of change in polar position with time for both regions rule out the possibility of a unique solution. It is therefore difficult to find an Eulerian pole of rotation describing the post-Carboniferous cumulative movement of the terrane, but we note tentatively that this pole could be near 30°N, 155°W, with a counterclockwise rotation angle of about 35°.

In terms of previously published hypotheses for the origin, movements, and history of the Alexander terrane, mentioned in the introduction, the third hypothesis of *Churkin and Eberlein* [1977] and also that of *Jones et al.* [1972] appear to be best supported by the paleomagnetic data. The hypothesis that the terrane is an ancient fragment of Asia or of an unknown and

TABLE 3. Plaeomagnetic Polar Positions for North American Craton

	Pole Identification (Reference)	Age	Polar Position	At 55.5°N, 133.5°W		At 40°N, 120°W	
				L	Decl	*L*	Decl
20	Late Permian mean [*Van der Voo and French,* 1974]	Pu	49°N, 113°E	28.0°N,	316.5	10.5°N,	328.0
19	Early Permian mean [*Van der Voo and French,* 1974]	Pl	43°N, 125°E	29.0°N,	305.0	11.5°N,	318.0
18	Late Carboniferous mean [*Van der Voo and French,* 1974]	Cu	40°N, 130°E	29.5°N,	300.0	12.0°N,	312.5
17	Mauch Chunk Fm. [*Knowles and Opdyke,* 1968]	Cl	43°N, 127°E	30.0°N,	304.0	12.5°N,	316.0
16	Catskill Fm., Penn. [*Van der Voo et al.,* 1979]	Du	45°N, 127°E	31.0°N,	305.5	14.0°N	318.0
15	Temple Butte Limestone [*Elston and Bressler,* 1977]	Dm-Du	53°N, 115°E	31.5°N	318.0	14.5°E	330.0
14	Martin Fm. [*Elston and Bressler,* [1977]	Dm-Du	56°N, 109°E	31.5°N	323.5	14.0°N	333.5
13	Catskill Fm., New York [*Kent and Opdyke,* 1978]	Dm-Du	47°N, 117°E	28.0°N	313.0	10.5°N	324.5
12	Delaware Limestone [*Martin,* 1975]	Dm	48°N, 118°E	30.0°N	317.0	12.0°N	324.5
11	Columbus Limestone [*Martin,* 1975]	Dm	45°N, 120°E	28.0°N	310.0	10.5°N	322.0
10	Bloomsburg Fm. [*Roy et al.,* 1967]	Su	32°N, 102°E	9.0°N	314.5	8.5°S	325.0
9	Rose Hill Fm. [*French and Van der Voo,* 1977]	Sm	26°N, 116°E	10.5°N	301.0	6.0°S	312.0
8	Chapman Ridge Fm. [*Watts and Van der Voo,* 1979]	(Om)Sl	27°N, 111°E	9.0°N	306.0	8.0°S	316.0
7	Beemerville complex [*Proko and Hargraves,* 1973]	Ou-Sl	35°N, 126°E	23.0°N	299.0	6.0°N	311.5
6	Juniata Fm. [*Van der Voo and French,* 1977]	Ou	32°N, 114°E	14.0°N	306.0	2.5°S	316.0
5	Trenton Limestone [*McElhinny and Opdyke,* 1973]	Om-Ou	36°N, 116°E	18.0°N	307.0	1.5°N	318.0
4	Moccasin/Bays Fm. [*Watts and Van der Voo,* 1979]	Om	33°N, 144°E	30.0°N	285.0	16.0°N	300.0
3	Wichita granite [*Spall,* 1970]	Cu	4°N, 164°E	18.0°N	249.0	12.5°N	264.0
2	Wilberns Fm. [*Van der Voo et al.,* 1976]	Cu	6°N, 159°E	17.5°N	255.0	10.5°N	270.0
1	Lamotte Fm. [*Al-Khafaji and Vincenz,* 1971]	Cu	1°N, 168°E	18.0°N	244.0	14.0°N	259.0

Explanation: Age: C–u = Late Cambrian, Om = Middle Ordovician, Ou = Late Ordovician, Sl = Early Silurian, Sm = Middle Silurian, Su = Late Silurian, Dm = Middle Devonian, Du = Late Devonian, Cl = early Carboniferous, Cu = late Carboniferous, Pl = Early Permian, Pu = Late Permian; *L* is ancient latitude at site (55.5°N, 133.5°W or 40°N, 120°W); Decl is predicted declination.

now-dispersed continent would require much larger movements in order to retain consistency with the observations. Accordion-like opening and closing of marginal seas, the second hypothesis summarized in the introduction, is not supported by the paleomagnetic results for the Paleozoic from Prince of Wales Island, assuming that these motions were roughly along lines of paleolatitude. Instead, a northward latitudinal movement of about 18° is associated with a counterclockwise rotation of at least 25° about a vertical axis through the sampling area. We note that this rotation brings the long axis of Prince of Wales Island roughly into alignment with the facies and tectonic trends of the western United States in Nevada and California. If the approximate paleoposition of 40°N, 120°W, is correct, it would place the Alexander terrane within the eugeosynclinal Paleozoic domain of northern California and northwestern Nevada, where other borderland terranes that are also probably allochthonous have been recognized [*Churkin and Eberlein,* 1977]. However, stratigraphic matching is difficult between the terranes of northern California–northwestern Nevada and the Alexander terrane, particularly for the basement. The Permian and Triassic period of extension in this area and other areas could well have initiated the breakaway and the relative north-northwestward movement. In a companion paper, *Hillhouse and Gromme* [1980] describe the paleomagnetic results from Triassic volcanic rocks at sites around Keku Strait (Kupreanof Island), 100 km north of our study area. Kupreanof Island is part of the Admiralty subterrane of the Alexander terrane [*Berg et al.,* 1978; *Hillhouse and Gromme,* 1980]. By Late Permian time the entire Alexander terrane behaved as one crustal unit, so that the results of Hillhouse and Gromme indicate that its northward drift was completed by the Late Triassic, although a counterclockwise post-Triassic rotation is still evident.

Paleomagnetic studies of other Mesozoic and Cenozoic rocks of the western belt of the North American Cordillera in general support similar northward motion of areas within the western belt with respect to the North American craton [*Beck,* 1976; *Hillhouse,* 1977*a*; *Jones et al.,* 1977]. The paleomagnetic results suggest motions ranging from 50° northward shifts to exclusively rotational motions about a vertical axis through the block involved. The sense of rotational motion is in general clockwise, with four notable exceptions: (1) the East Sooke Gabbro of Vancouver Island [*Symons,* 1973], (2) the Triassic Nikolai Greenstone of Alaska [*Hillhouse,* 1977*a*], (3) the Triassic volcanic rocks of Keku Strait [*Hillhouse and Gromme,* 1980], and (4) the rocks in our study. The Nikolai Greenstone could have been rotated clockwise, but the amount of northward translation would then increase from 27° to 57°. A clockwise rotation of the Prince of Wales Island area is likewise undesirable because of the unlikely increase in both northward movement and rotation about a vertical axis (at least 145°). *Beck*'s [1976] hypothesis of a ball-bearing type mechanism for translation of small microplates caught between the North American plate and the ancestral Pacific plate(s), with a resulting northward translation and a clockwise sense of motion, apparently does not hold for all the microplates.

Because of the different amounts of northward motion and the different senses of rotation concluded from paleomagnetic studies of Paleozoic, Mesozoic, and Cenozoic rocks, the ancient borderland terranes within the western or outer belt of the Cordillera appear to be composed of microplates. This interpretation is further substantiated by the differences in stratigraphy and particularly in the type of basement of the various terranes within the belt [*Churkin and Eberlein,* 1977].

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